

Introduction and Objective of the Study:

The goal of this Exploratory Data Analysis (EDA) is to identify which factors influence the number of car seat units sold (Sales) at each store location. The dataset contains both numerical predictors (prices, income, age, etc.) and categorical predictors (Shelving Quality, Urban vs Rural, US vs Non-US). The analysis follows a fully structured approach combining visual exploration, statistical tests, and regression modelling.

The workflow is as follows:

1. Data Check and Cleaning **2. Exploratory Visualisations and Descriptive Analysis** >> **3. Correlation and Association Analysis** >> **4. Statistical Tests** >> **5. Regression Analysis** >> **6. Summary of Interpretation and Insights**

Each step of the workflow is detailed below.

Importing Libraries

```
In [6]: import pandas as pd
import matplotlib.pyplot as plt
import numpy as np
import seaborn as sns
from scipy import stats
import statsmodels.api as sm
from statsmodels.stats.outliers_influence import variance_inflation_factor
from scipy.stats import ttest_ind
from scipy.stats import ttest_rel
import warnings
from statsmodels.formula.api import ols
```

```
In [7]: from bokeh.layouts import row
from bokeh.models import ColumnDataSource, DataTable, TableColumn
from bokeh.plotting import figure, show
from bokeh.layouts import gridplot
import holoviews as hv
```

Step 1. Data Check and Cleaning

The dataset is generally clean, with 400 records and only 4 instances containing at least one missing value. Even though this represents just 1% of the dataset and the missing values are spread across different variables, we have chosen not to keep or impute them - rather delete the records so there will not be any issue while doing ANOVA or Regression.

The dataset contains two main of variable types:

Numeric: Sales, CompPrice, Income, Advertising, Price, Age (continuous); Population and Education (discrete)

Categorical: ShelveLoc (Shelving Quality), Urban (Yes/No), US (Yes/No)

```
In [8]: # Loading the dataset
df_main = pd.read_excel("CarSeats.xlsx")
```

```
In [9]: # List of variables and their data type
df_main.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 400 entries, 0 to 399
Data columns (total 11 columns):
#   Column      Non-Null Count  Dtype
---  -
0   Sales       399 non-null    float64
1   CompPrice    399 non-null    float64
2   Income       399 non-null    float64
3   Advertising  400 non-null    int64
4   Population   400 non-null    int64
5   Price        400 non-null    int64
6   ShelveLoc    400 non-null    object
7   Age          400 non-null    int64
8   Education    400 non-null    int64
9   Urban        399 non-null    object
10  US           400 non-null    object
dtypes: float64(3), int64(5), object(3)
memory usage: 34.5+ KB
```

```
In [10]: # A snippet into the first 5 rows of the dataset
df_main.head()
```

```
Out[10]:
```

	Sales	CompPrice	Income	Advertising	Population	Price	ShelveLoc	Age	Education	Urban	US
0	9.50	138.0	73.0	11	276	120	Bad	42	17	Yes	Yes
1	11.22	111.0	48.0	16	260	83	Good	65	10	Yes	Yes
2	10.06	113.0	35.0	10	269	80	Medium	59	12	Yes	Yes
3	7.40	117.0	100.0	4	466	97	Medium	55	14	Yes	Yes
4	4.15	141.0	64.0	3	340	128	Bad	38	13	Yes	No

```
In [11]: # A quick sum of the missing values in each variable
df_main.isnull().sum()
```

```
Out[11]: Sales      1
CompPrice  1
Income     1
Advertising 0
Population 0
Price      0
ShelveLoc  0
Age        0
Education  0
Urban      1
US         0
dtype: int64
```

```
In [12]: # The records containing the null values
df_main[df_main[['Sales', 'CompPrice', 'Income', 'Urban']].isna().any(axis=1)]
```

```
Out[12]:
```

	Sales	CompPrice	Income	Advertising	Population	Price	ShelveLoc	Age	Education	Urban	US
19	8.73	129.0	NaN	16	58	121	Medium	69	12	Yes	Yes
22	5.08	NaN	46.0	6	497	138	Medium	42	13	Yes	No
51	NaN	121.0	90.0	0	150	108	Bad	75	16	Yes	No
62	1.82	139.0	45.0	0	146	133	Bad	77	17	NaN	Yes

```
In [13]: # Deleting the records with null values to avoid errors in ANOVA or other statistical analyses
df = df_main.replace([np.inf, -np.inf], np.nan).dropna()
df.isna().sum()
```

```
Out[13]: Sales      0
CompPrice  0
Income     0
Advertising 0
Population 0
Price      0
ShelveLoc  0
Age        0
Education  0
Urban      0
US         0
dtype: int64
```

Step 2. Exploratory Visualisations and Descriptive Analysis

In this section we are exploring the distribution of the variables in the dataset using descriptive statistics and visualisations like histogram and barplot. It shows that several numeric variables (Sales, Advertising, Population) have high variance and skewness, which is explored further through visualizations to assess distributions and potential outliers.

For categorical values - first we checked the unique values of each variables and its count before plotting them in a bar chart. They are not evenly distributed as the charts show. Exploratory Visualisations in this section lead the way to further exploration in the Correlation and Association section.

Numeric Variables: Distributions and Outliers

```
In [9]: # A basic summary statistics of the numerical variables of the dataset
df.describe()
```

Out[9]:

	Sales	CompPrice	Income	Advertising	Population	Price	Age	Education
count	396.000000	396.000000	396.000000	396.000000	396.000000	396.000000	396.000000	396.000000
mean	7.521414	124.931818	68.70202	6.646465	265.366162	115.702020	53.196970	13.893939
std	2.816309	15.392350	28.05604	6.650498	147.056382	23.749368	16.172068	2.624879
min	0.000000	77.000000	21.00000	0.000000	10.000000	24.000000	25.000000	10.000000
25%	5.457500	115.000000	42.00000	0.000000	139.000000	100.000000	39.000000	12.000000
50%	7.495000	124.500000	69.00000	5.000000	274.000000	117.000000	54.000000	14.000000
75%	9.322500	135.000000	91.00000	12.000000	398.500000	131.000000	65.250000	16.000000
max	16.270000	175.000000	120.00000	29.000000	509.000000	191.000000	80.000000	18.000000

```
In [10]: #List of numeric columns
num_cols = ['Sales', 'CompPrice', 'Income', 'Advertising', 'Population', 'Price', 'Age', 'Education']

#Column Labels
ncol_names = ['Units Sold (in thousands)', 'Competitor Price', 'Community Income', 'Advertising Cost', 'Population (in thousand',

# 2 rows - for distribution and histogram, 8 columns for 8 variables
fig, axes = plt.subplots(2, 8, figsize=(24, 8))

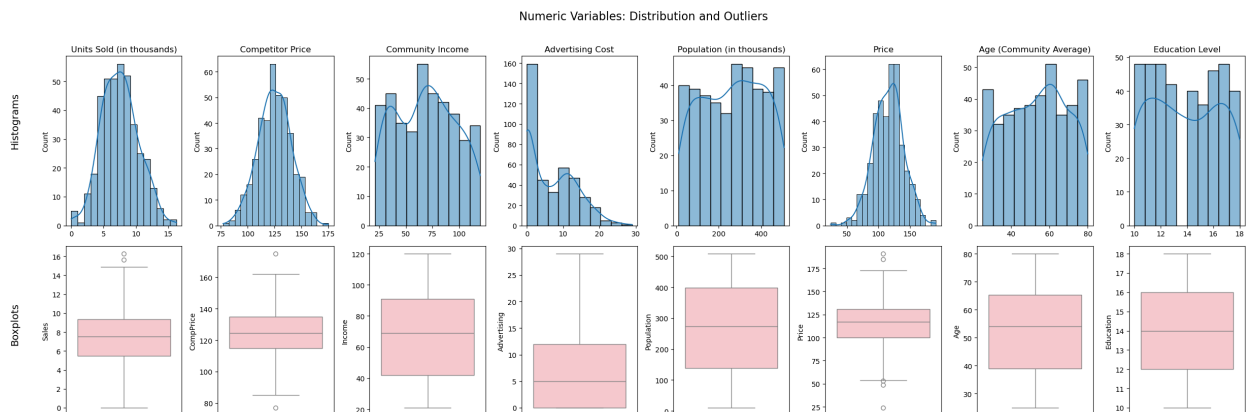
for i, (col, name) in enumerate(zip(num_cols, ncol_names)):
    # Histogram in first row
    sns.histplot(df[col], kde=True, ax=axes[0, i])
    axes[0, i].set_title(f'{name}')
    axes[0, i].set_xlabel('') # Removing x-axis titles to declutter the charts

    # Boxplot in second row
    sns.boxplot(y=df[col], ax=axes[1, i], color='pink')

# Adding a single title for the whole grid
fig.suptitle('Numeric Variables: Distribution and Outliers', fontsize=16, y=1.02)

# Adding row Labels (y-position adjusted)
fig.text(-0.02, 0.75, 'Histograms', fontsize=14, va='center', rotation='vertical')
fig.text(-0.02, 0.25, 'Boxplots', fontsize=14, va='center', rotation='vertical')

plt.tight_layout()
plt.show()
```



Key Takeaways: Units sold, price of car seats, and competitor price are roughly normally distributed with minor skew. A few locations show extreme values, such as very high sales or unusually low/high prices, but these outliers are rare. Advertising cost is heavily left skewed suggesting that most locations had a small advertising budget. Other variables like Population, Age, and Income have higher variance and mild skew, reflecting heterogeneity among locations. These patterns will be explored further in the correlation and statistical testing sections.

Categorical Variables: Distributions

```
In [15]: from bokeh.io import output_notebook
output_notebook()

cat_cols = ['ShelveLoc', 'Urban', 'US'] # categorical columns
ccol_names = ['Shelf Quality', 'Urban Store', 'US Store'] # corresponding display names

plots = []
tables = []

for col, display_name in zip(cat_cols, ccol_names):
    vc = df[col].value_counts(dropna=False)
    values = [str(v) for v in vc.index.tolist()]
```

```
counts = vc.values

src = ColumnDataSource(data=dict(
    value=values,
    count=counts
))

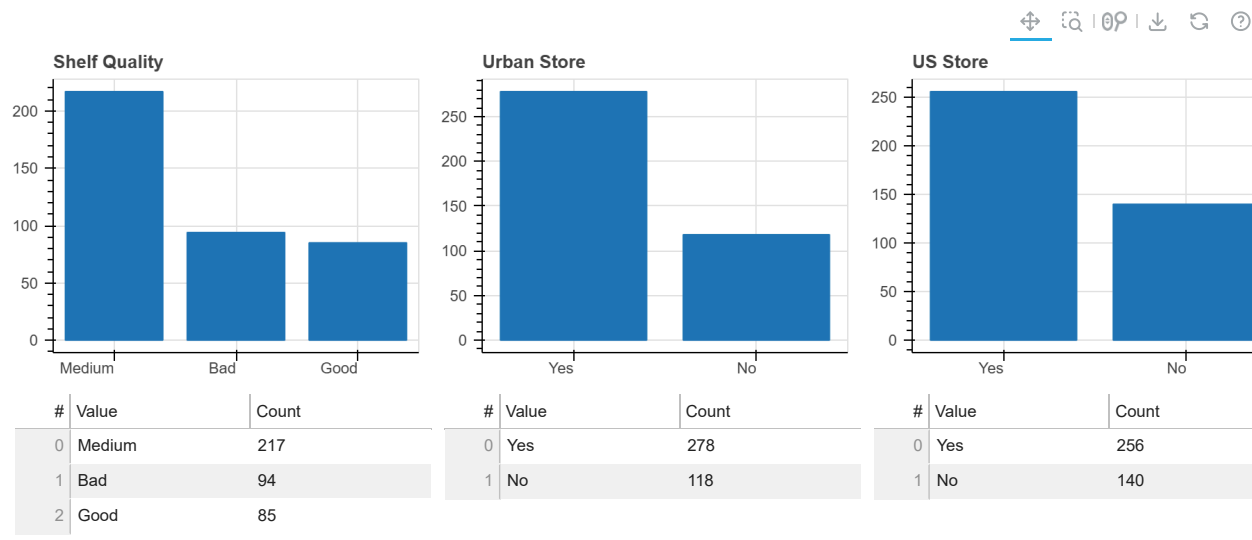
# Bar chart
p = figure(
    x_range=values,
    height=250, width=300,
    title=f"{display_name}",
    toolbar_location=None
)
p.vbar(x="value", top="count", width=0.8, source=src)
p.xaxis.major_label_orientation = 0 # no rotation

# Table
columns = [
    TableColumn(field="value", title="Value"),
    TableColumn(field="count", title="Count"),
]
tbl = DataTable(source=src, columns=columns, width=300, height=350)

plots.append(p)
tables.append(tbl)

# Arrange first row with all charts, second row with all tables
grid = gridplot([plots, tables])
show(grid)
```

Loading BokehJS ...



Key Takeaways: Most of the products have the Shelf Position of Medium quality. The stores are mostly situated in urban area and almost 2/3rd of the shops are situated in the US.

```
In [17]: med_income = df['Income'].median()

subset1 = df[(df['Income'] >= med_income) & (df['Urban'] == 'Yes')]
subset1.describe()
```

	Sales	CompPrice	Income	Advertising	Population	Price	Age	Education
count	147.000000	147.000000	147.000000	147.000000	147.000000	147.000000	147.000000	147.000000
mean	7.824286	123.632653	91.292517	6.741497	262.863946	113.605442	54.176871	13.680272
std	2.900506	16.499930	14.928235	6.570053	149.172587	24.653366	15.384880	2.671479
min	0.530000	77.000000	69.000000	0.000000	13.000000	24.000000	25.000000	10.000000
25%	5.865000	114.000000	80.000000	0.000000	135.500000	98.000000	42.000000	11.000000
50%	7.540000	123.000000	89.000000	5.000000	262.000000	116.000000	56.000000	13.000000
75%	10.055000	135.000000	102.000000	12.000000	403.000000	129.500000	65.500000	16.000000
max	14.370000	161.000000	120.000000	25.000000	508.000000	166.000000	80.000000	18.000000

Step 3: Correlation and Association Analysis

From the distribution patterns observed in the previous section, it appears that several variables may be correlated and could influence sales. In this section, we explore these relationships using visualizations, a Pearson correlation matrix for numeric variables, and Chi-square tests for categorical variables. Furthermore, based on the insights generated from the exploratory visualisations, we delved deeper and look for any correlation between the categorical attributes of the store and the number of sales.

Correlation Matrix for Numerical Variables

Are there any correlation between the sales and the other numerical attributes of the store and the product?

Numerical attributes e.g. competitor price, product price, income of the community, population in the community, etc.

```
In [18]: #dataframe with numerical values only
df_n = df.loc[:, df.dtypes != 'object']
df_n.head()
```

	Sales	CompPrice	Income	Advertising	Population	Price	Age	Education
0	9.50	138.0	73.0	11	276	120	42	17
1	11.22	111.0	48.0	16	260	83	65	10
2	10.06	113.0	35.0	10	269	80	59	12
3	7.40	117.0	100.0	4	466	97	55	14
4	4.15	141.0	64.0	3	340	128	38	13

```
In [19]: #creating the correlation matrix
correlation_matrix = df_n.corr()
correlation_matrix
```

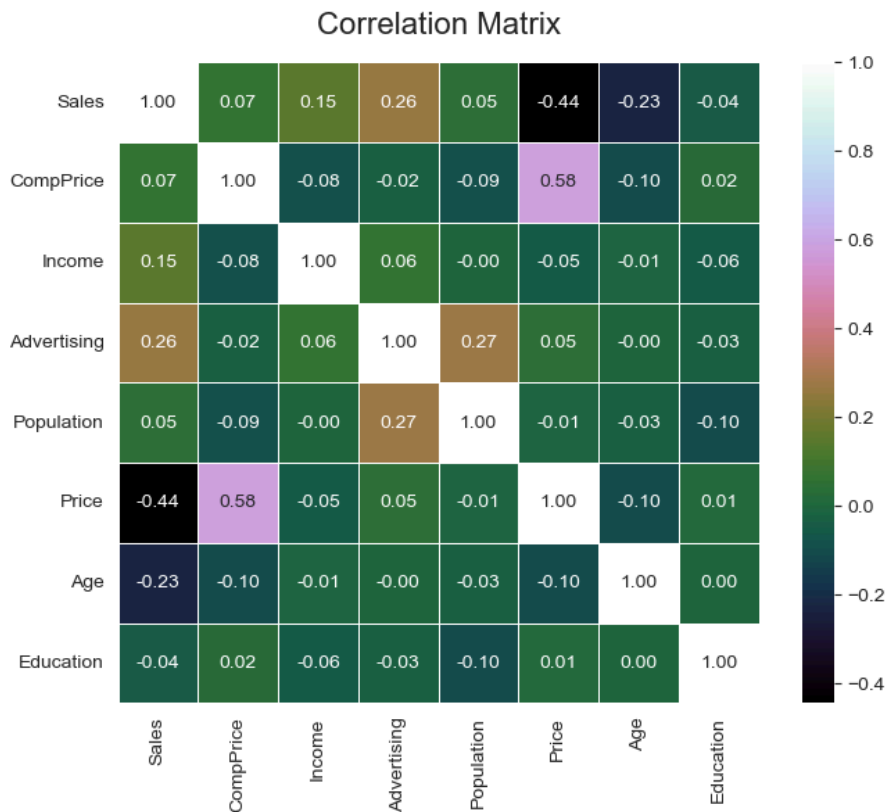
	Sales	CompPrice	Income	Advertising	Population	Price	Age	Education
Sales	1.000000	0.068760	0.149274	0.263351	0.049909	-0.444711	-0.226571	-0.044222
CompPrice	0.068760	1.000000	-0.078286	-0.023582	-0.094004	0.584242	-0.103866	0.023757
Income	0.149274	-0.078286	1.000000	0.058103	-0.004021	-0.053003	-0.006231	-0.056430
Advertising	0.263351	-0.023582	0.058103	1.000000	0.270297	0.045301	-0.001140	-0.026372
Population	0.049909	-0.094004	-0.004021	0.270297	1.000000	-0.014365	-0.031341	-0.104712
Price	-0.444711	0.584242	-0.053003	0.045301	-0.014365	1.000000	-0.103558	0.011513
Age	-0.226571	-0.103866	-0.006231	-0.001140	-0.031341	-0.103558	1.000000	0.000613
Education	-0.044222	0.023757	-0.056430	-0.026372	-0.104712	0.011513	0.000613	1.000000

```
In [20]: #creating heatmap from the correlation matrix

plt.figure(figsize=(8,6))
sns.set_style("whitegrid")

# Heatmap
sns.heatmap(correlation_matrix, annot=True, fmt=".2f", cmap="cubehelix", cbar=True, square=True, linewidths=0.5)

plt.title("Correlation Matrix", fontsize=16, pad=15)
plt.show()
```



Key Takeaways from Correlation Matrix

Price is the strongest factor negatively affecting sales.

Advertising has some positive effect, while demographic variables (Population, Age, Education) have minimal influence.

Competitor price is not a major driver in this dataset. But the matrix gives us the insight that stores with higher prices tend to have competitors with higher prices too.

Are there any correlation between the sales and the store's geographical location as well as the car seat's shelf position?

```
In [22]: # Getting all categorical columns
cat_cols = [col for col in df.columns if df[col].dtype == 'object']
col_names = ['Shelf Quality', 'Urban Store', 'US Store']

sns.set_style("whitegrid")
palette = "Set2" # soft colors
# Creating subplots: 1 row, as many columns as categorical variables
fig, axes = plt.subplots(1, len(cat_cols), figsize=(5 * len(cat_cols), 5), squeeze=False)

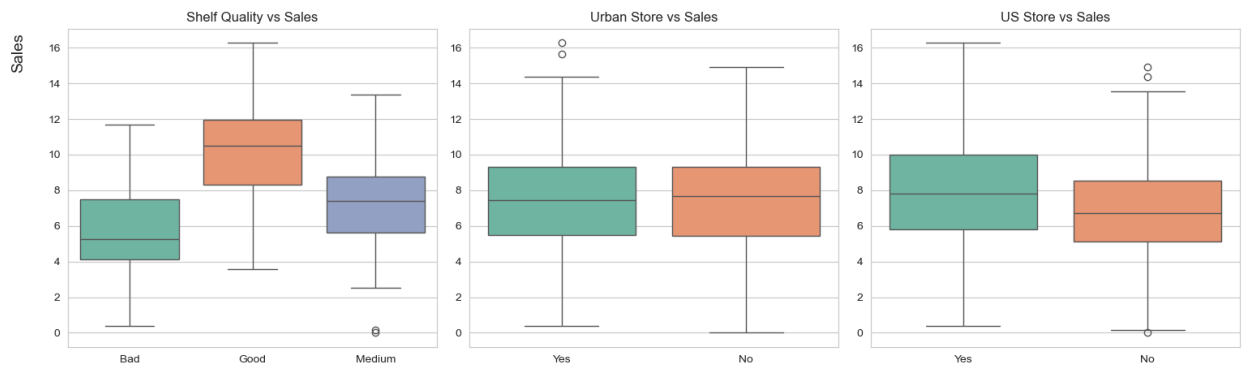
for i, (col, name) in enumerate(zip(cat_cols, col_names)):
    sns.boxplot(x=col, y='Sales', data=df, ax=axes[0, i], palette = palette)
    axes[0, i].set_title(f"{name} vs Sales")
    axes[0, i].tick_params(axis='x')
    axes[0, i].set_xlabel('') # decluttering by removing axis titles
    axes[0, i].set_ylabel('')

# Adding a single title for the whole grid
fig.suptitle('How is Sales based on its Shelf Location and Geographic Location?', fontsize=16, y=1.02)
# Adding row labels (y-position adjusted)
fig.text(-0.02, 0.75, 'Sales', fontsize=14, va='bottom', rotation = 'vertical')

plt.tight_layout()
plt.show()

warnings.filterwarnings('ignore')
```

How is Sales based on its Shelf Location and Geographic Location?



Key Takeaways from Visualisation

Shelf Quality: Medium shelf locations are most common but show the widest price variation. Some Medium-quality stores have sales lower than those with Bad shelf quality. Good shelf quality shows the highest median sales.

Urban vs Non-Urban: Urban stores are twice as many, with a few very high performers. However, Non-Urban stores show stronger consistency, achieving a similar median sales level despite being fewer.

US vs Non-US: Non-US stores have high outliers, while US stores show a wider sales range overall.

```
In [23]: # Set a clean style
sns.set_style("whitegrid")
palette = "Set3"
fig, axes = plt.subplots(1, 2, figsize=(14, 5))

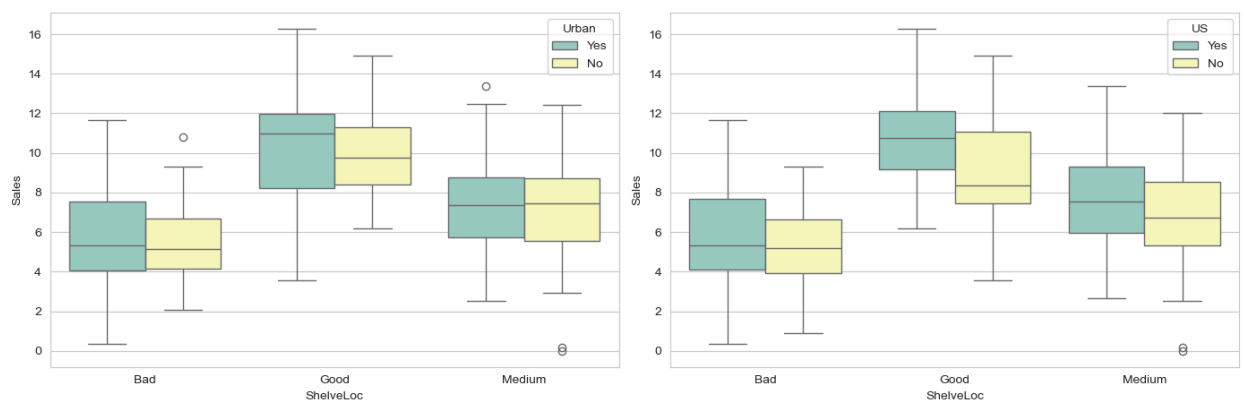
# Plot 1: ShelfLoc x Sales x Urban
sns.boxplot(data=df, x="ShelfLoc", y="Sales", hue="Urban", ax=axes[0], palette = palette)

# Plot 2: ShelfLoc x Sales x US
sns.boxplot(data=df, x="ShelfLoc", y="Sales", hue="US", ax=axes[1], palette = palette)

# Adding a single title for the whole grid
fig.suptitle('A Side-by-side Comparison of Sales by Shelf Position and Geographic Location', fontsize=16, y=1.02)

plt.tight_layout()
plt.show()
```

A Side-by-side Comparison of Sales by Shelf Position and Geographic Location



Key Takeaways from the charts

As seen in the previous chart, Urban and non-Urban stores have similar sales. But, stores in the US have overall higher sales than non-US ones. Some of the stores where the Car Seats are positioned in a Medium quality shelf location have worse sales in the non-urban and non-US area. Urban and US stores have better consistency in this regard. But even though there are not many non-US stores - it is worthwhile to mention that non-US markets had greater variability in performance. This variability suggests there may be underlying market differences worth exploring further.

Chi-square Test

Shelf Position and Store Location Correlation Statistical Test

Can we say whether the Shelf position good bad or medium is correlated to whether the store is in the US or in Urban area? We need to do a chi-square test for that. The hypotheses are:

Hypothesis 1

H_0 : Shelf position (Good, Medium, Bad) is independent of whether a store is Urban or Non-Urban.

H_A : Shelf position is dependent on whether a store is Urban or Non-Urban.

Hypothesis 2

H_0 : Shelf position (Good, Medium, Bad) is independent of whether a store is in US.

H_A : Shelf position is dependent on whether a store is in US.

```
In [27]: # To test Hypothesis 1, creating contingency table
cont_table1 = pd.crosstab(df['ShelveLoc'], df['Urban'])
cont_table1
```

```
Out[27]:
```

	Urban	No	Yes
ShelveLoc			
Bad	22	72	
Good	28	57	
Medium	68	149	

```
In [28]: # Performing the test
chi2, p_val, dof, expected = stats.chi2_contingency(cont_table1)
print(f"p-value: {p_val}")
```

p-value: 0.28881804548584356

```
In [29]: # To test Hypothesis 2, creating contingency table
cont_table2 = pd.crosstab(df['ShelveLoc'], df['US'])
cont_table2
```

```
Out[29]:
```

	US	No	Yes
ShelveLoc			
Bad	33	61	
Good	24	61	
Medium	83	134	

```
In [30]: # Performing the test
chi2, p_val, dof, expected = stats.chi2_contingency(cont_table2)
print(f"p-value: {p_val}")
```

p-value: 0.2614665318248534

Key Takeaway from Chi-Square Test

We failed to reject both the hypotheses. Car Seat's shelf position quality has no significant correlation to the store's location whether US or Urban.

Anova Test

In the Pearson correlation matrix and chi-square tests, we first examine pairwise relationships: numeric variables with Sales, and categorical variables like Shelf Location with US/Urban. For instance, Price shows a moderate negative correlation with Sales (~ -0.45), while Advertising shows a weak positive correlation (~ 0.27). Shelf position shows no correlation.

Pearson's correlation and Chi-square are limited because it is pairwise, linear, and cannot test independence between numeric and categorical variables. To address this, we use ANOVA, which allows us to assess the effect of multiple predictors (numeric and categorical) on Sales simultaneously, determine which variables significantly contribute, and compare the magnitude of effects using F-statistics and p-values.

The Hypotheses are:

Hypothesis 1:

H_0 : Mean Sales is the same for all Shelf Locations (Good, Medium, Bad)

H_A : Mean Sales differs for at least one Shelf Location

Hypotheses 2,...,7:

H_0 : The predictor has no effect on Sales (coefficient = 0)

H_A : The predictor has a significant effect on Sales (coefficient $\neq 0$)

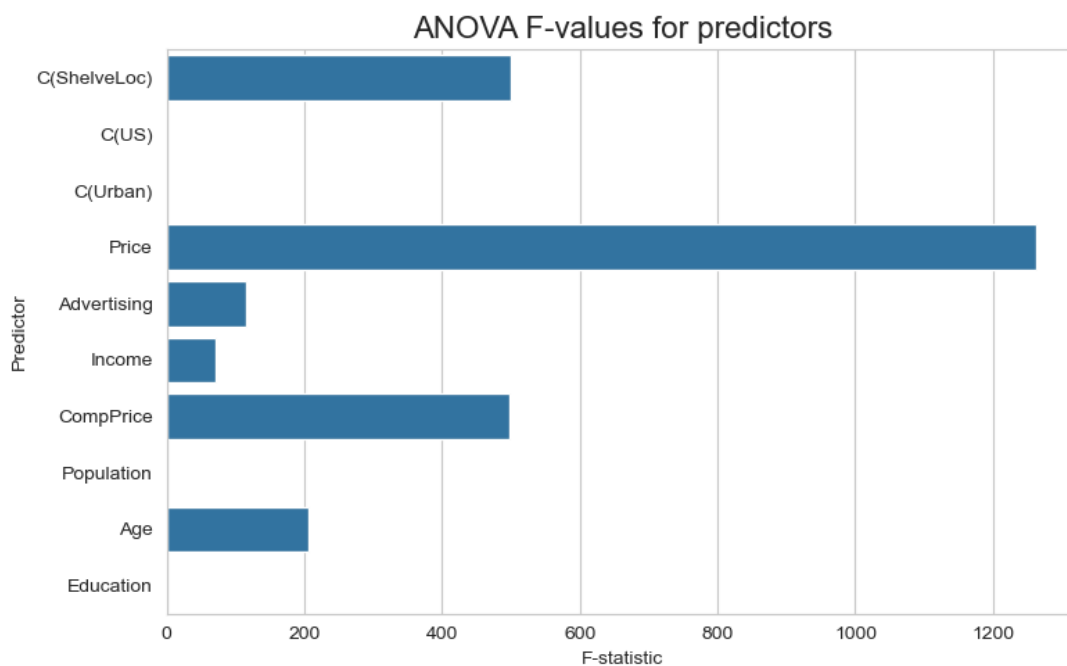
Here the predictors are Price, Advertising Cost, Community Income, Competitor Price, Community Population, Average Age of Community, Community Education Level.

```
In [31]: # Linear model with categorical and numeric predictors -
# we are including almost all variables here even though some of their pairwise correlation was low.
model_full = ols('Sales ~ C(ShelveLoc) + C(US) + C(Urban) + Price + Advertising + Income + CompPrice + Population + Age + Education', data=train)
anova = sm.stats.anova_lm(model_full, typ=2)
print(anova)
```

	sum_sq	df	F	PR(>F)
C(ShelveLoc)	1042.304709	2.0	499.614938	1.379996e-107
C(US)	1.382549	1.0	1.325413	2.503392e-01
C(Urban)	1.287569	1.0	1.234358	2.672563e-01
Price	1316.288369	1.0	1261.890744	2.039117e-123
Advertising	121.646993	1.0	116.619745	6.431234e-24
Income	74.536837	1.0	71.456489	5.965347e-16
CompPrice	519.389784	1.0	497.925208	2.520345e-71
Population	0.507686	1.0	0.486706	4.858228e-01
Age	214.409964	1.0	205.549145	1.224529e-37
Education	1.004652	1.0	0.963133	3.270167e-01
Residual	400.553484	384.0	NaN	NaN

```
In [32]: # Preparing ANOVA table for visualisation (excluding residuals)
anova_plot = anova.reset_index().rename(columns={'index': 'Variable'})
anova_plot = anova_plot[anova_plot['Variable'] != 'Residual'] # drop residual

plt.figure(figsize=(8,5))
sns.barplot(x='F', y='Variable', data=anova_plot)
plt.title('ANOVA F-values for predictors', fontsize=16)
plt.xlabel('F-statistic')
plt.ylabel('Predictor')
plt.tight_layout()
plt.show()
```



```
In [33]: # Confidence Interval. If a predictor's confidence interval does not include 0 the model is significant.
# Even though we got F and p value to measure a predictor's explanation power of variance in the dependent variable -
# - and p-value to understand the significance of the F-value I am still calculating the CI
# CI gives us the range of values in which the true effect of the predictor is likely to lie.

print(model_full.conf_int(alpha=0.05)) # 95% CI
```

	0	1
Intercept	4.437511	6.820657
C(ShelveLoc)[T.Good]	4.539635	5.145975
C(ShelveLoc)[T.Medium]	1.697319	2.198791
C(US)[T.Yes]	-0.472631	0.123546
C(Urban)[T.Yes]	-0.096996	0.349032
Price	-0.100508	-0.089965
Advertising	0.099360	0.143595
Income	0.012026	0.019315
CompPrice	0.084639	0.100996
Population	-0.000475	0.000997
Age	-0.052367	-0.039736
Education	-0.058475	0.019536

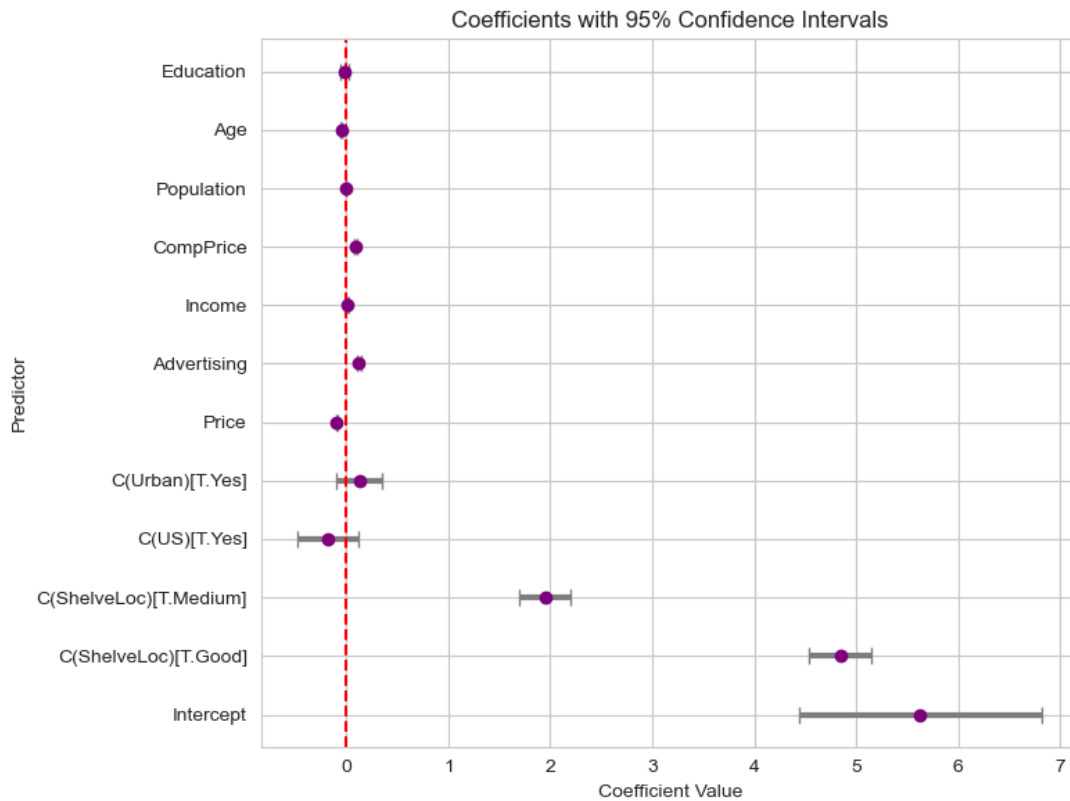
```
In [37]: # Getting coefficients and CI
coefs = model_full.params
```

```

conf = model_full.conf_int()
conf['coef'] = coefs
conf.columns = ['CI_lower', 'CI_upper', 'coef']
conf = conf.reset_index().rename(columns={'index': 'Variable'})

# Plot
plt.figure(figsize=(8,6))
plt.errorbar(x=conf['coef'], y=conf['Variable'],
             xerr=[conf['CI_lower'] - conf['coef'], conf['CI_upper'] - conf['coef']],
             fmt='o', color='purple', ecolor='grey', elinewidth=3, capsize=4)
plt.axvline(x=0, color='red', linestyle='--') # zero-effect line
plt.title('Coefficients with 95% Confidence Intervals')
plt.xlabel('Coefficient Value')
plt.ylabel('Predictor')
plt.tight_layout()
plt.show()

```



In [38]: *# Is there multi-collinearity? Computing VIF (Variance Inflation Factor) to test multi-collinearity*

```

X = df[['Price', 'Advertising', 'Income', 'CompPrice', 'Population', 'Age', 'Education']]
X = sm.add_constant(X)
vif = pd.DataFrame()
vif['Variable'] = X.columns
vif['VIF'] = [variance_inflation_factor(X.values, i) for i in range(X.shape[1])]
print(vif)

```

	Variable	VIF
0	const	133.588758
1	Price	1.532236
2	Advertising	1.087370
3	Income	1.013808
4	CompPrice	1.547467
5	Population	1.102918
6	Age	1.015699
7	Education	1.014531

Key Takeaways from ANOVA

All the hypotheses are rejected except Community Education and Population. Meaning Shelf Postion and the other predictors except Community Education and Population have a significant effect on the Sales - corroborated by the F value, p value and the Confidence Intervals.

Significant F-values and very small p-values indicate that all the variable has a meaningful effect on Sales, even if its simple correlation with Sales is low. For example, Income may have a low correlation with Sales individually, but in the multivariable ANOVA model, it explains part of the variance after accounting for other variables, resulting in a significant F-value.

All predictors have VIF values below 2 (except the intercept), indicating low multicollinearity. This suggests that the effects of individual variables on Sales are reliably estimated and the ANOVA results are stable.

More ANOVA

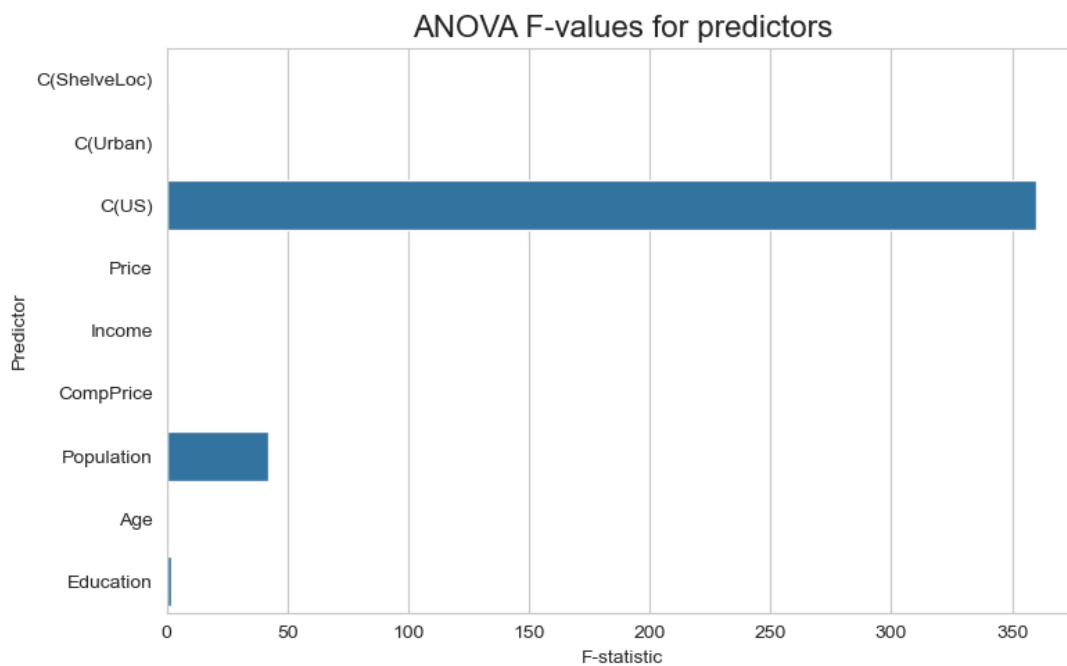
This is time we are trying to see if there are any significant predictor behind Advertisement cost. The aim is to combine the finding with the sales predictors and develop a deeper understanding of the car seat sales pattern.

```
In [39]: model_adv = ols('Advertising ~ C(ShelveLoc) + C(Urban) + C(US) + Price + Income + CompPrice + Population + Age + Education')
anova_adv = sm.stats.anova_lm(model_adv, typ=2)
print(anova_adv)
```

	sum_sq	df	F	PR(>F)
C(ShelveLoc)	29.404633	2.0	0.686652	5.038736e-01
C(Urban)	12.900353	1.0	0.602493	4.381049e-01
C(US)	7704.174570	1.0	359.812889	3.961210e-57
Price	6.258658	1.0	0.292302	5.890617e-01
Income	0.025239	1.0	0.001179	9.726292e-01
CompPrice	10.127070	1.0	0.472971	4.920377e-01
Population	899.673656	1.0	42.018022	2.773264e-10
Age	0.083910	1.0	0.003919	9.501165e-01
Education	46.041417	1.0	2.150301	1.433582e-01
Residual	8243.471260	385.0	NaN	NaN

```
In [41]: # Preparing ANOVA table for visualisation (excluding residuals)
anova_plot_adv = anova_adv.reset_index().rename(columns={'index': 'Variable'})
anova_plot_adv = anova_plot_adv[anova_plot_adv['Variable'] != 'Residual'] # drop residual

plt.figure(figsize=(8,5))
sns.barplot(x='F', y='Variable', data=anova_plot_adv)
plt.title('ANOVA F-values for predictors', fontsize=16)
plt.xlabel('F-statistic')
plt.ylabel('Predictor')
plt.tight_layout()
plt.show()
```



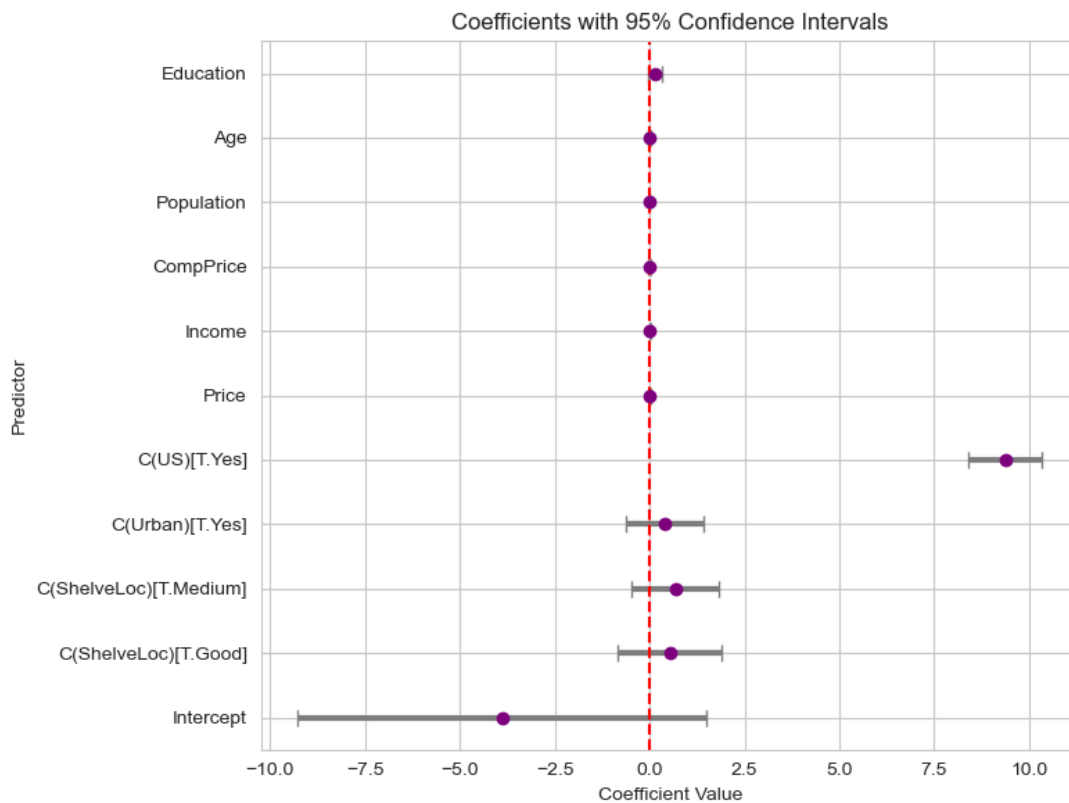
```
In [29]: print(model_adv.conf_int(alpha=0.05)) # 95% CI
```

	0	1
Intercept	-9.263941	1.505133
C(ShelveLoc)[T.Good]	-0.840255	1.904764
C(ShelveLoc)[T.Medium]	-0.460929	1.807036
C(Urban)[T.Yes]	-0.611025	1.408174
C(US)[T.Yes]	8.396676	10.338626
Price	-0.017308	0.030437
Income	-0.016225	0.016802
CompPrice	-0.049983	0.024078
Population	0.007268	0.013596
Age	-0.029524	0.027702
Education	-0.044794	0.307663

```
In [42]: # Getting coefficients and CI
coefs = model_adv.params
conf = model_adv.conf_int()
conf['coef'] = coefs
conf.columns = ['CI_lower', 'CI_upper', 'coef']
conf = conf.reset_index().rename(columns={'index': 'Variable'})

# Plot
```

```
plt.figure(figsize=(8,6))
plt.errorbar(x=conf['coef'], y=conf['Variable'],
            xerr=[conf['coef'] - conf['CI_lower'], conf['CI_upper'] - conf['coef']],
            fmt='o', color='purple', ecolor='grey', elinewidth=3, capsize=4)
plt.axvline(x=0, color='red', linestyle='--') # zero-effect line
plt.title('Coefficients with 95% Confidence Intervals')
plt.xlabel('Coefficient Value')
plt.ylabel('Predictor')
plt.tight_layout()
plt.show()
```



Key Takeaways from ANOVA

Advertisement cost is high in US and in denser population area. Education also has a significant role but the confidence interval contains 0. So it is not as significant as US and population variable.

Step 4: Statistical Tests

In the previous section we looked into the pair-wise correlation between the variables. With ANOVA we also tried finding any significant predictors to Sales volume. In this section we will conduct independent two sample t-tests to gain further insight into the store's location, area demography and product attributes for a better market understanding. The questions can be:

Question 1: Do Urban stores have high income customers as expected?

$H_0 : \mu_1 = \mu_2$; There is no income difference of the community whether the store is in Urban or in Rural area $H_A : \mu_1 > \mu_2$; The community income level in Urban is higher in the Urban than in the Rural area

Question 2: Do we have high sales in the Urban area?

$H_0 : \mu_1 = \mu_2$; Sales is not different between Urban and Rural area $H_A : \mu_1 > \mu_2$; Sales is higher in the Urban than in the Rural area

Question 3: Do the car seats cost higher in the Urban area? $H_0 : \mu_1 = \mu_2$; Car Seats cost the same in Urban and Rural area $H_A : \mu_1 > \mu_2$; Car Seats cost is higher in Urban than in Rural area

Question 4: Do we have higher revenue in the Urban area? $H_0 : \mu_1 = \mu_2$; Revenue from selling Car Seats is the same in the Rural and the Urban area $H_A : \mu_1 > \mu_2$; Revenue from selling Car Seats is higher in the Urban than in the Rural area

Question 5: Is Price different than Competitor's Price? $H_0 : \mu_1 = \mu_2$; Price is not different than Competitor's Price $H_A : \mu_1 < \mu_2$; Price is lower than Competitor's Price

```
In [30]: # Question 1: Do Urban stores have high income customers as expected?
# High Income ( $\geq$  median) & Urban stores
```

```
urban_income = df[df['Urban'] == 'Yes']['Income']
rural_income = df[df['Urban'] == 'No']['Income']
urban_income.describe(), rural_income.describe()
```

```
Out[30]: (count      278.000000
mean       69.417266
std        27.361175
min        21.000000
25%        44.000000
50%        70.000000
75%        91.000000
max       120.000000
Name: Income, dtype: float64,
count      118.000000
mean       67.016949
std        29.680919
min        21.000000
25%        40.250000
50%        67.000000
75%        90.000000
max       120.000000
Name: Income, dtype: float64)
```

```
In [31]: # The variance is different. So running Welch's t-test
t_stat, p_two_tailed = ttest_ind(urban_income, rural_income, equal_var=False)

# If group1 mean > group2 mean
if t_stat > 0:
    p_one_tailed = p_two_tailed / 2
else:
    p_one_tailed = 1 - (p_two_tailed / 2)

print("t-statistic:", t_stat)
print("one-tailed p-value:", p_one_tailed)
```

```
t-statistic: 0.7530959128914143
one-tailed p-value: 0.22612741218121446
```

The one-tailed Welch's t-test shows that Urban stores do not have significantly higher community income than Rural stores ($t = 0.75$, $p = 0.23$). Therefore, we cannot conclude that Urban locations serve higher-income communities.

```
In [32]: # Question 2: Do we have high sales in the Urban area?
```

```
urban_sales = df[df['Urban'] == 'Yes']['Sales']
rural_sales = df[df['Urban'] == 'No']['Sales']
urban_sales.describe(), rural_sales.describe()
```

```
Out[32]: (count      278.000000
mean       7.503525
std        2.825597
min        0.370000
25%        5.482500
50%        7.435000
75%        9.317500
max       16.270000
Name: Sales, dtype: float64,
count      118.000000
mean       7.563559
std        2.805846
min        0.000000
25%        5.440000
50%        7.670000
75%        9.335000
max       14.900000
Name: Sales, dtype: float64)
```

```
In [33]: # The variance is slightly different. So running Welch's t-test
t_stat, p_two_tailed = ttest_ind(urban_sales, rural_sales, equal_var=False)

# If group1 mean > group2 mean
if t_stat > 0:
    p_one_tailed = p_two_tailed / 2
else:
    p_one_tailed = 1 - (p_two_tailed / 2)

print("t-statistic:", t_stat)
print("one-tailed p-value:", p_one_tailed)
```

```
t-statistic: -0.19432917997828536
one-tailed p-value: 0.5769520875285206
```

The t-statistic is negative, meaning the mean of Urban sales is actually slightly lower than Rural sales. But with p value greater than 0.05, there is no statistically significant evidence that Urban stores have higher sales than Rural stores. The observed difference in means could

easily be due to random variation.

```
In [34]: # Question 3: Do the car seats cost higher in the Urban area?
```

```
urban_price = df[df['Urban'] == 'Yes']['Price']
rural_price = df[df['Urban'] == 'No']['Price']
urban_price.describe(), rural_price.describe()
```

```
Out[34]: (count      278.000000
         mean      116.392086
         std       24.052910
         min       24.000000
         25%      101.000000
         50%      118.000000
         75%      131.000000
         max      191.000000
         Name: Price, dtype: float64,
         count      118.000000
         mean      114.076271
         std       23.037004
         min       49.000000
         25%       97.000000
         50%      112.500000
         75%      128.000000
         max      185.000000
         Name: Price, dtype: float64)
```

```
In [35]: # The variance is slightly different. So running Welch's t-test
t_stat, p_two_tailed = ttest_ind(urban_price, rural_price, equal_var=False)

# If group1 mean > group2 mean
if t_stat > 0:
    p_one_tailed = p_two_tailed / 2
else:
    p_one_tailed = 1 - (p_two_tailed / 2)

print("t-statistic:", t_stat)
print("one-tailed p-value:", p_one_tailed)
```

```
t-statistic: 0.902896470486678
one-tailed p-value: 0.1837636235623616
```

There is no statistically significant evidence that Urban stores charge higher prices than Rural stores. The observed difference could easily be due to random variation.

```
In [36]: # Question 4: Do we have higher revenue in the Urban area?
```

```
df['Revenue'] = df['Sales']*df['Price'] # creating a new column Revenue = Sales*Price

urban_revenue = df[df['Urban'] == 'Yes']['Revenue']
rural_revenue = df[df['Urban'] == 'No']['Revenue']
urban_revenue.describe(), rural_revenue.describe()
```

```
Out[36]: (count      278.000000
         mean      844.916331
         std      309.174515
         min       70.670000
         25%      622.260000
         50%      824.300000
         75%     1025.797500
         max     1794.260000
         Name: Revenue, dtype: float64,
         count      118.000000
         mean      830.341017
         std      284.685306
         min        0.000000
         25%      652.540000
         50%      844.645000
         75%      994.532500
         max     1586.230000
         Name: Revenue, dtype: float64)
```

```
In [37]: # The variance is slightly different. So running Welch's t-test
t_stat, p_two_tailed = ttest_ind(urban_revenue, rural_revenue, equal_var=False)

# If group1 mean > group2 mean
if t_stat > 0:
    p_one_tailed = p_two_tailed / 2
else:
    p_one_tailed = 1 - (p_two_tailed / 2)

print("t-statistic:", t_stat)
print("one-tailed p-value:", p_one_tailed)
```

t-statistic: 0.4540016771435566
one-tailed p-value: 0.3251204602387639

Again, there is no statistically significant evidence that Urban stores charge higher prices than Rural stores. The observed difference could easily be due to random variation.

In [38]: # Question 5: Is Price different than Competitor's Price?

```
t_stat, p_val = ttest_rel(df['Price'], df['CompPrice'])

if t_stat < 0:
    p_one_tailed = p_val / 2
else:
    p_one_tailed = 1 - (p_val / 2)

print("t-statistic:", t_stat)
print("one-tailed p-value:", p_one_tailed)
```

t-statistic: -9.499824842111856
one-tailed p-value: 1.0309998015761473e-19

We reject the null hypothesis. Our Car Seat price is lower compared to the Competitor's Price.

Step 5: Regression Analysis

We are going to perform a multiple regression analysis where Sales is the dependent variable and independent variables will include all variables proven statistically significant predictors in the ANOVA analysis earlier.

$$Sales = \alpha + \beta_1 * ShelfPosition + \beta_2 * Price + \beta_3 * AdvertisingCost + \beta_4 * Income + \beta_5 * CompetitorPrice + \beta_6 * Age$$

In [41]: model = ols('Sales ~ C(ShelveLoc) + Price + Advertising + Income + CompPrice + Age', data=df).fit()

In [42]: model.summary()

Out[42]:

OLS Regression Results						
Dep. Variable:	Sales	R-squared:	0.871			
Model:	OLS	Adj. R-squared:	0.868			
Method:	Least Squares	F-statistic:	373.3			
Date:	Thu, 11 Dec 2025	Prob (F-statistic):	4.92e-168			
Time:	16:26:59	Log-Likelihood:	-566.36			
No. Observations:	396	AIC:	1149.			
Df Residuals:	388	BIC:	1181.			
Df Model:	7					
Covariance Type:	nonrobust					
	coef	std err	t	P> t	[0.025	0.975]
Intercept	5.4845	0.507	10.818	0.000	4.488	6.481
C(ShelveLoc)[T.Good]	4.8256	0.154	31.416	0.000	4.524	5.128
C(ShelveLoc)[T.Medium]	1.9390	0.127	15.288	0.000	1.690	2.188
Price	-0.0952	0.003	-35.524	0.000	-0.100	-0.090
Advertising	0.1150	0.008	14.788	0.000	0.100	0.130
Income	0.0156	0.002	8.476	0.000	0.012	0.019
CompPrice	0.0926	0.004	22.382	0.000	0.084	0.101
Age	-0.0461	0.003	-14.363	0.000	-0.052	-0.040
Omnibus:	0.729	Durbin-Watson:	1.989			
Prob(Omnibus):	0.694	Jarque-Bera (JB):	0.781			
Skew:	0.102	Prob(JB):	0.677			
Kurtosis:	2.923	Cond. No.	1.91e+03			

Notes:

[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.

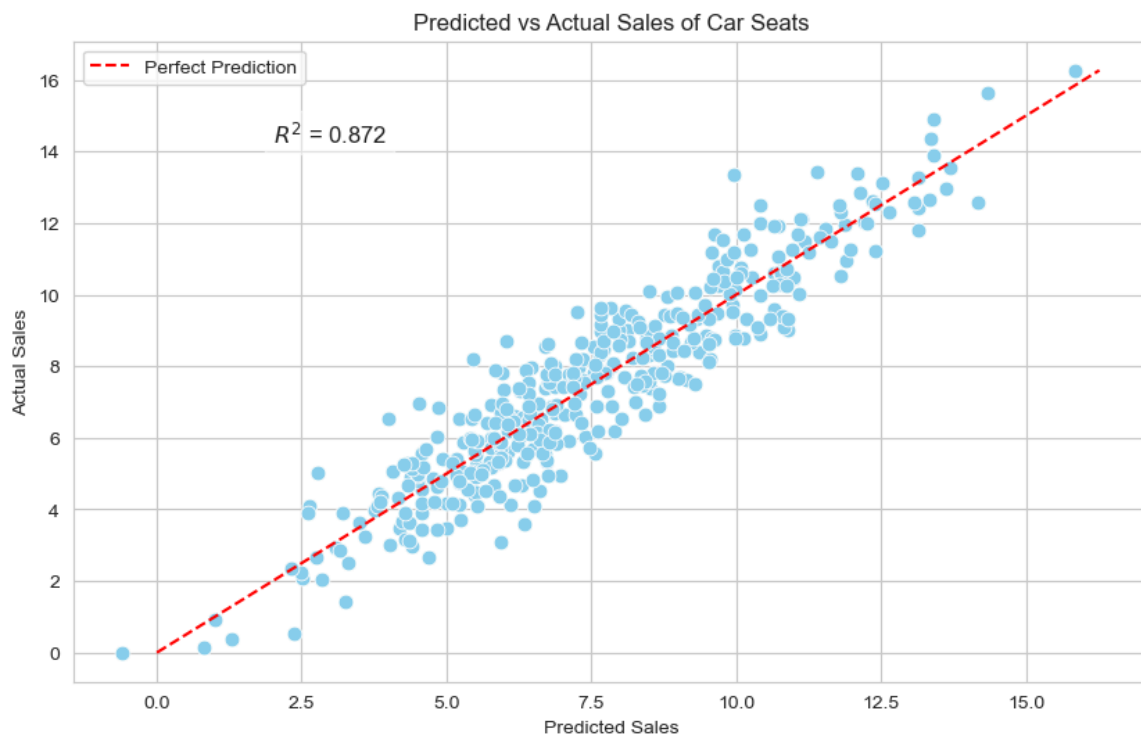
[2] The condition number is large, 1.91e+03. This might indicate that there are strong multicollinearity or other numerical problems.

```
In [43]: import matplotlib.pyplot as plt
import seaborn as sns

# Predicted values
df['Predicted_Sales'] = model_full.fittedvalues

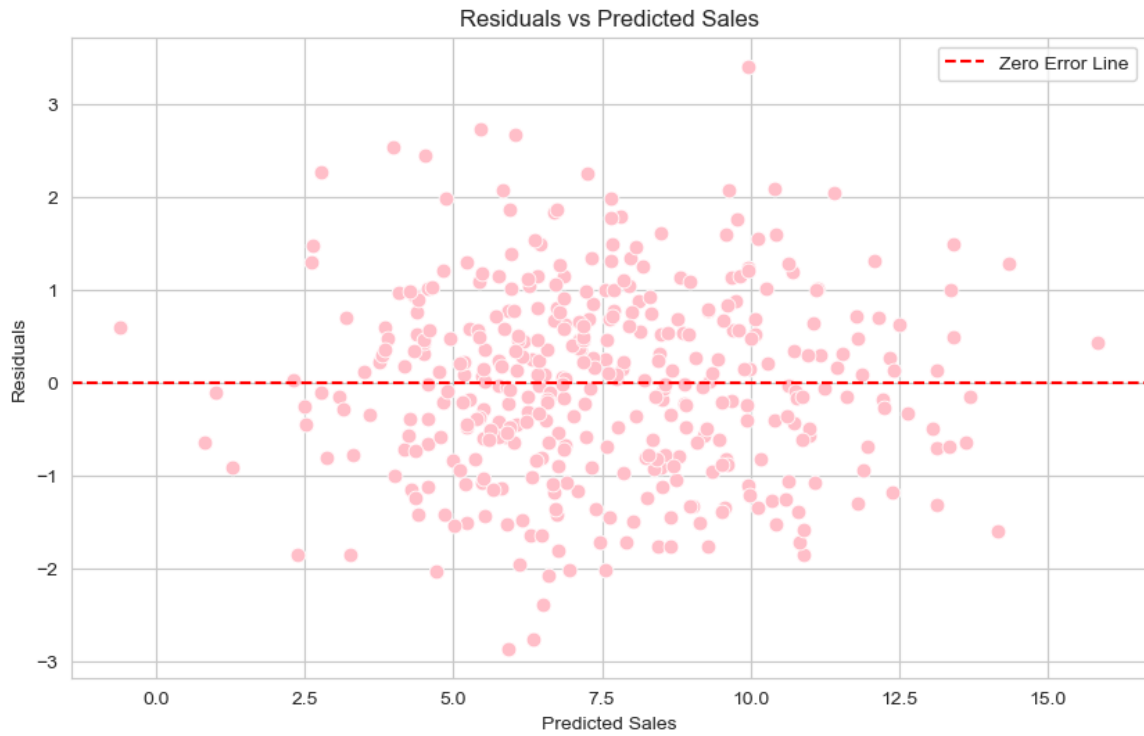
# R-squared
r2 = model_full.rsquared

# Plot
plt.figure(figsize=(10,6))
sns.scatterplot(x='Predicted_Sales', y='Sales', data=df, color='skyblue', s=60)
plt.plot([df['Sales'].min(), df['Sales'].max()],
         [df['Sales'].min(), df['Sales'].max()],
         color='red', linestyle='--', label='Perfect Prediction')
plt.text(df['Sales'].min()+2, df['Sales'].max()-2, f'$R^2$ = {r2:.3f}',
        fontsize=12, bbox=dict(facecolor='white', alpha=0.5))
plt.title('Predicted vs Actual Sales of Car Seats')
plt.xlabel('Predicted Sales')
plt.ylabel('Actual Sales')
plt.grid(True)
plt.legend()
plt.show()
```

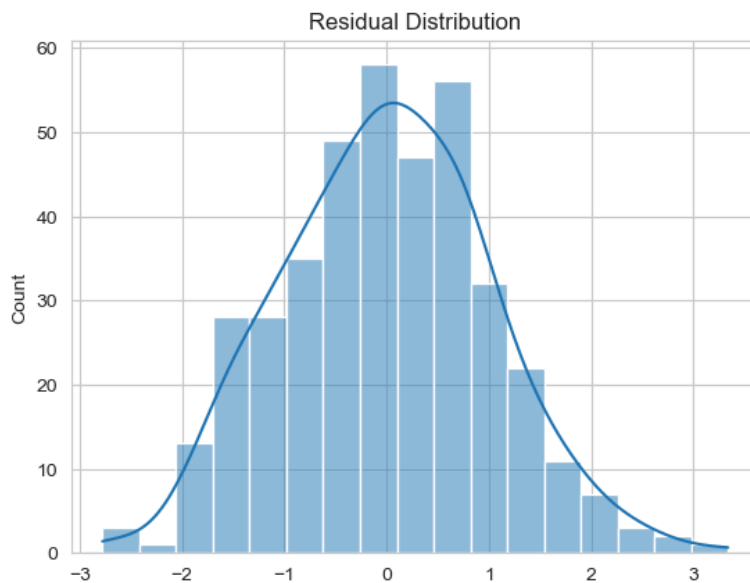


```
In [44]: # Calculate residuals
df['Residuals'] = df['Sales'] - df['Predicted_Sales']

# Residuals vs Predicted plot
plt.figure(figsize=(10,6))
sns.scatterplot(x='Predicted_Sales', y='Residuals', data=df, color='pink', s=60)
plt.axhline(0, color='red', linestyle='--', label='Zero Error Line')
plt.title('Residuals vs Predicted Sales')
plt.xlabel('Predicted Sales')
plt.ylabel('Residuals')
plt.grid(True)
plt.legend()
plt.show()
```

```
In [45]: # checking residual distribution
resid = model.resid
sns.histplot(resid, kde=True)
plt.title("Residual Distribution")
plt.show()
```



Key Takeaways

The OLS regression explains 87% of the variance in Sales ($R^2 = 0.871$).

Shelf location, Price, Advertising, Income, CompPrice, and Age are all statistically significant predictors ($p < 0.001$).

Positive coefficients for Good/Medium shelf location, Advertising, Income, and CompPrice indicate higher Sales, while Price and Age negatively impact Sales.

Jarque Bera test affirms the normality of the data distribution. The Residual Distribution is also normal - thus all the assumptions of the multiple linear regression test are proved.

Step 6: Summary of Interpretation and Insights

The analysis of car seat sales across locations reveals several key patterns. From the descriptive statistics and visualizations, numeric variables like Sales, Advertising, and Population exhibit high variance and some skewness, indicating that a few locations have unusually high sales, advertising spend, or population. Categorical variables such as Shelf Location, Urban, and US show clear group differences in Sales.

Correlation analysis shows moderate negative correlation between Price and Sales and weak positive correlations between Advertising, Income, and Sales. Shelf Location has little pairwise correlation with Sales but is highly significant in ANOVA and regression, highlighting its categorical effect. T-tests comparing Urban vs. Rural areas indicate that Urban stores tend to have slightly higher Income, Sales, and Price, although the differences are not statistically significant at all, while paired t-tests show that Price is consistently lower than Competitor Price across locations. Advertisement Cost is more in the US and where the population is higher.

The multiple regression model confirms these patterns: Shelf Location, Price, Advertising, Income, Competitor Price, and Age all significantly influence Sales. Good shelving positions, higher advertising spend, higher competitor prices, and higher community income increase Sales, while higher Price and older population reduce it. Multicollinearity is mild. Residual diagnostics suggest that regression assumptions hold reasonably well.

Overall, the analysis highlights that **shelf positioning, pricing strategy, local demography** and **promotional spending** are key drivers of car seat sales. These insights provide actionable guidance for optimizing store layout, pricing, and promotional strategies to increase revenue.

Recommendations

1. Shelf Placement Optimization

Observation: Shelf Location is the strongest categorical predictor of Sales. **Recommendation:** Prioritize placing car seats in “Good” shelving locations to maximize visibility and sales. Medium locations are also significantly better than “Bad,” so improving shelf quality can yield incremental gains.

2. Pricing Strategy

Observation: Price negatively affects sales, showing customer sensitivity, while Competitor Price positively affects demand.

Recommendation: Consider pricing competitively but avoid underpricing too much — ensure that your price positioning considers competitor pricing. Dynamic pricing or localized pricing strategies may help maximize revenue.

3. Advertising Allocation

Observation: Advertising increases Sales and is correlated with Population and US locations, but correlation with Sales is modest.

Recommendation: Allocate advertising strategically, focusing on areas where the impact on sales is expected to be higher (e.g., high-population areas, locations with “Good” shelf placement).

4. Target High-Income Areas

Observation: Income positively affects Sales. **Recommendation:** Tailor promotions or premium product options in higher-income regions to capture higher demand.

5. Consider Demographic Effects

Observation: Age has a negative association with Sales. **Recommendation:** Focus marketing and placement strategies in areas with younger populations, or introduce complementary products for older demographics if targeting them.

6. Monitor Outliers and High-Leverage Locations

Observation: Some locations have unusually high Sales, Advertising, or Population. **Recommendation:** Identify these outliers to understand best practices or special circumstances — e.g., high-performing stores might indicate successful layouts, promotions, or market characteristics worth replicating elsewhere.

7. Continuous Monitoring

Implement a dashboard to track Sales, Advertising, and Price effects over time and by location. This allows quick adjustment to marketing, pricing, and inventory strategies.

Acknowledgement

AI tools were used to support understanding and learning of relevant programming (e.g. chart creation in bokeh, using grid, troubleshooting etc) and data analysis techniques.